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Pontryagin's minimum principle-based real-time energy management strategy for fuel cell hybrid electric vehicle considering both fuel economy and power source durability

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ABSTRACT

This paper presents a real-time and approximately optimal energy management strategy (EMS) based on Pontryagin's minimum principle (PMP), considering both fuel economy and power source durability. To develop the target strategy, performance degradation models are built for two power sources: a proton exchange membrane fuel cell and a lithium ion battery. This study provides an online co-state updating method for uncertain driving cycles. The method allows the battery's state of charge to be controlled within a certain range and determines the nearly optimal hydrogen consumption. Furthermore, by incorporating a fuel cell power variation limiting factor with a weight coefficient into the PMP to suppress power changes, the durability of the fuel cell can be improved. The average daily operating cost (calculated based on the fuel consumption and power source degradation) is used to evaluate the trade-off between fuel economy and power source durability. Simulation results show that the fuel cell durability is improved with a slight increase in fuel consumption and battery degradation, and that the average daily operating cost is effectively reduced. A comparison with the results obtained by adopting a rule-based EMS and a dynamic programming-based EMS indicate the superiority of the proposed EMS.

1. Introduction

Increases in environmental pollution and shortages of fossil fuels have accelerated the commercialisation of clean energy vehicles [1]. A fuel cell has advantages in regards to high efficiency, zero emissions, a short refuelling time, and a long driving range, and is considered as the most promising power source for clean energy vehicles [2–4]. Owing to the poor dynamic responses of proton exchange membrane fuel cells (PEMFCs), the power systems of fuel cell hybrid electric vehicles (FCHEVs) are often equipped with an auxiliary energy source, such as a battery or super capacitor. The auxiliary energy source plays a key role in providing a fast transient response and braking energy recovery, and enhances the vehicle's dynamic performance and powertrain efficiency [5–7].

For a fuel cell/battery hybrid power system, the mission of an energy management strategy (EMS) is to allocate the power demand between the two power sources. The EMS attempts to achieve a system-level performance objective while satisfying powertrain component constraints [8]. EMSs based on global optimisation algorithms such as dynamic programming (DP) and Pontryagin's minimum principle (PMP) can determine an optimal allocation strategy for a specific vehicle under a known driving cycle. The PMP algorithm is an optimal control theory that can be used to determine the optimal solution for a dynamic control system and is especially applicable to constrained systems. In earlier studies of PMP-based EMSs for PEMFCs, the optimisation goal was to minimise the fuel consumption, the state variable was the battery state of charge (SOC), and the control variable was the battery power or fuel cell system (FCS) net power [9–12]. Compared with the DP algorithm, the PMP algorithm transforms a global optimisation problem into an instantaneous optimisation problem by introducing a co-state, providing the possibility for online

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Nomenclatures

Acronyms

EMS	Energy Management Strategy
FCHEV	Fuel Cell Hybrid Electric Vehicle
PEMFC	Proton Exchange Membrane Fuel Cell
PMP	Pontryagin's Minimum Principle
SOC	State Of Charge
DP	Dynamic Programming
FCS	Fuel Cell System
HWFET	High Way Fuel Economy Test
UDDS	Urban Dynamometer Driving Schedule
US06	The US06 Supplemental Federal Test Procedure

Roman symbols

C_D	Coefficient of air resistance
A	Front area[m^2]
α	Inclination angle of the road[rad]
u	Vehicle speed[m s^{-1}]
f	Coefficient of rolling resistance
δ	The correction coefficient of rotating mass
F_t	Traction force[N]
T_w	The torque on the wheel[N m]
w_w	The wheel rotation speed[rad s^{-1}]
T_m	Torque on the motor shaft[N m]
w_m	Rotation speed of motor shaft[rad s^{-1}]
η_m	Motor efficiency
R_{fd}	The final drive gear ratio
P_t	Demanded traction power[kW]
P_{fc}	Power of fuel cell[kW]
P_{bat}	Power of battery[kW]
P_{req}	Total demanded power[kW]
M_{H_2}	Hydrogen consumption[g]
E_{low,H_2}	Low heating value of hydrogen [J g^{-1}]
D_{fc}	Degradation of fuel cell
d_{start_stop}	Fuel cell degradation caused by start/stop cycles
d_{low}	Fuel cell degradation caused by idling/low power conditions
d_{load_change}	Fuel cell degradation caused by load change cycles
d_{high}	Fuel cell degradation caused by high power conditions
$Signal_{engine\_on}$	Power on' signal of the FCS at the nth time step
P_{low}	Low-power threshold of fuel cell output power [kW]
P_{high}	High-power threshold of fuel cell output power [kW]
U_{oc}	Open circuit voltage of battery [V]
I_{bat}	Current of battery [A]
R_{bat}	Resistance of battery [Ω]
C_{bat}	Capacity of battery [Ah]
D_{bat}	Degradation of battery
T	Temperature [K]
a,b,c,d,e,f	Fitting coefficients
J	Performance Index function [g]
$\dot{m}$	Hydrogen consumption rate [g s^{-1}]
H	Hamiltonian function
p	Weight coefficient of power variation limiting factor
Std	Average of the standard deviation of fuel cell power
E	Energy capacity of the battery [kWh]
m_e	Equivalent hydrogen consumption [g]
M	Total hydrogen consumption [g]
c_{fc}	Price of the fuel cell [$ kW^{-1}]
c_{bat}	Price of the fuel cell [$ kWh^{-1}]
c_{H_2}	Price of hydrogen [$ kg^{-1}]
$Cost$	Average daily operating cost [$ day^{-1}]

Greek symbols

$\eta_{DC/DC}$	Efficiency of high voltage DC/DC converter
η_{fc}	Efficiency of fuel cell
η_c	Coulomb efficiency of battery
λ	Co-state
Δt	time step size [s]

implementation of PMP [13]. However, the initial value of the co-state is related to prior knowledge of the driving cycles, limiting online application. Using velocity prediction to update the co-state is an effective method for achieving approximately optimal control. Li [14] proposed an adaptive PMP-based EMS for FCHEVs. In this strategy, the co-state can be updated according to a future velocity prediction. This is a quasi-optimal strategy, as the optimal horizon is the prediction horizon instead of the entire driving cycle, but is also very computationally intensive. The influence of the prediction horizon length on the computational time and fuel economy was examined in Ref. [15]; a relatively short prediction horizon can appropriately balance the fuel economy and computational load. Onori [16] proposed another method for updating the co-state. The total driving distance and average vehicle velocity (e.g. as obtained from a GPS) are used to determine the initial value of co-state, and the SOC of the battery is used to update the co-state during driving. The performance of the strategy depends on the accuracy of the travel information prediction. The EMSs mentioned above can achieve satisfactory fuel economy and have the potential for real-time application. However, none of them consider fuel cell durability.

Fuel cell durability is a technical challenge that must be overcome in the commercialisation of FCHEVs. In automotive applications, the degradation of a fuel cell is mainly caused by frequent changes in the FCS output power [1,17]. Therefore, developing EMSs based on frequency separation is an effective approach. With frequency separation methods [18–20] such as wavelet transforms and low-pass filters, the low-frequency components of the demanded power are allocated to the PEMFC, and the remaining high-frequency components are allocated to auxiliary power sources. However, frequency separation-based EMSs only focus on the durability of the fuel cell system, and may lead to high hydrogen consumption. Another feasible method is applied when designing optimisation-based EMSs for minimum hydrogen consumption. A penalty term for power changes is added into a cost function to suppress the power changes in the fuel cells, so as to improve their respective lifetimes [21–23]. Owing to the trade-off between fuel economy and fuel cell durability, the weighting factor for the penalty term requires careful choice.

In view of the above, it is essential to build an effective real-time EMS for improving fuel economy and prolonging the lifetime of fuel cells. To realise an online implementation of PMP, the initial value of the co-state should be reasonably determined, and an effective updating method should be provided. Prolonging the lifetime of the fuel cell will lead to a higher fuel consumption; thus, this trade-off must be balanced. Furthermore, the smooth output power of the fuel cell will lead to worse operating conditions for the battery, causing a rapid degradation of the battery. Thus, not only the durability of the fuel cell, but also that of the battery should be considered.

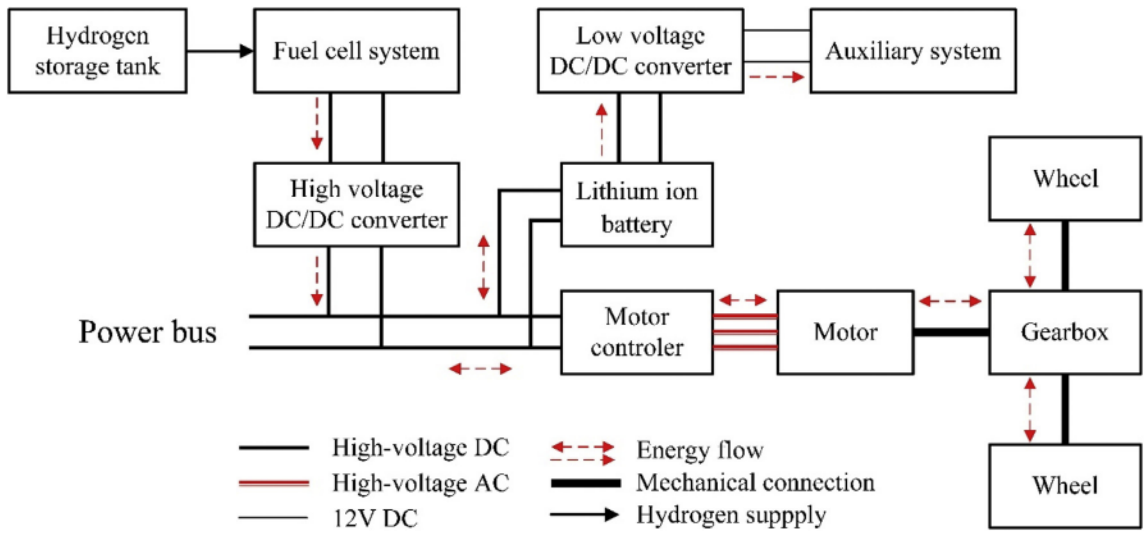

Fig. 1. Structure of the fuel cell-battery hybrid electric vehicle powertrain.

In this paper, a real-time PMP-based EMS is proposed for a fuel cell/battery hybrid passenger car, considering both fuel consumption and power source durability. In Section 2, the structure of the fuel cell powertrain and system model are introduced, including degradation models for the fuel cell and lithium ion battery. In Section 3, a novel EMS is designed, based on using the relationship between co-states and SOCs to realise an online implementation of PMP. Furthermore, a fuel cell power variation limiting factor with a weight coefficient is introduced to the proposed EMS, so as to consider power source durability. Simulation results from the proposed EMS and two other EMSs are compared in Section 4. Finally, conclusions are provided in Section 5.

2. Powertrain structure and model

Fig. 1 illustrates the structure of the fuel cell/battery hybrid electric powertrain, consisting of a PEMFC, high voltage DC/DC converter, lithium ion battery package, low voltage DC/DC converter, permanent magnet synchronous motor, and gearbox. Table 1 lists the key parameters of the FCHEV investigated in this study.

2.1. Vehicle modelling

According to the longitudinal dynamics of the vehicle, the traction force is calculated as follows:

$$F_t = mgf\cos\alpha + \frac{C_D A u^2}{21.15} + mg\sin\alpha + \delta m\frac{du}{dt} \tag{1}$$

The torque on the wheel T_w and wheel rotation speed w_w are then calculated as follows:

$$\begin{aligned} T_w &= rF_t \\ w_w &= u/r \end{aligned} \tag{2}$$

The torque T_m and the rotational speed w_m on the motor shaft are calculated as follows:

Table 1
Parameters of the fuel cell hybrid electric vehicles (FCHEV).

		Unit	Value
Vehicle	Vehicle weight	kg	2064(unloaded)/2244(loaded)
	Front area	m^2	2.588
	Wheel radius	m	0.347
	Coefficient of air resistance	–	0.375
	Coefficient of rolling resistance	–	0.009
	Final drive ratio	–	8.867
Fuel cell system (FCS)	Maximal power	kW	82(Gross output) 65(Net output)
	Effective area of cell	cm^2	270
	Number of cells	–	480
	DC/DC efficiency	–	0.97
Battery	Capacity	Ah	37
	Rated voltage	V	350
	Allowed range of state of charge (SOC)	%	5–95
Motor	Maximal power	kW	105
	Rated/maximal speed	rpm	3907/12000
	Maximal torque	N m	250

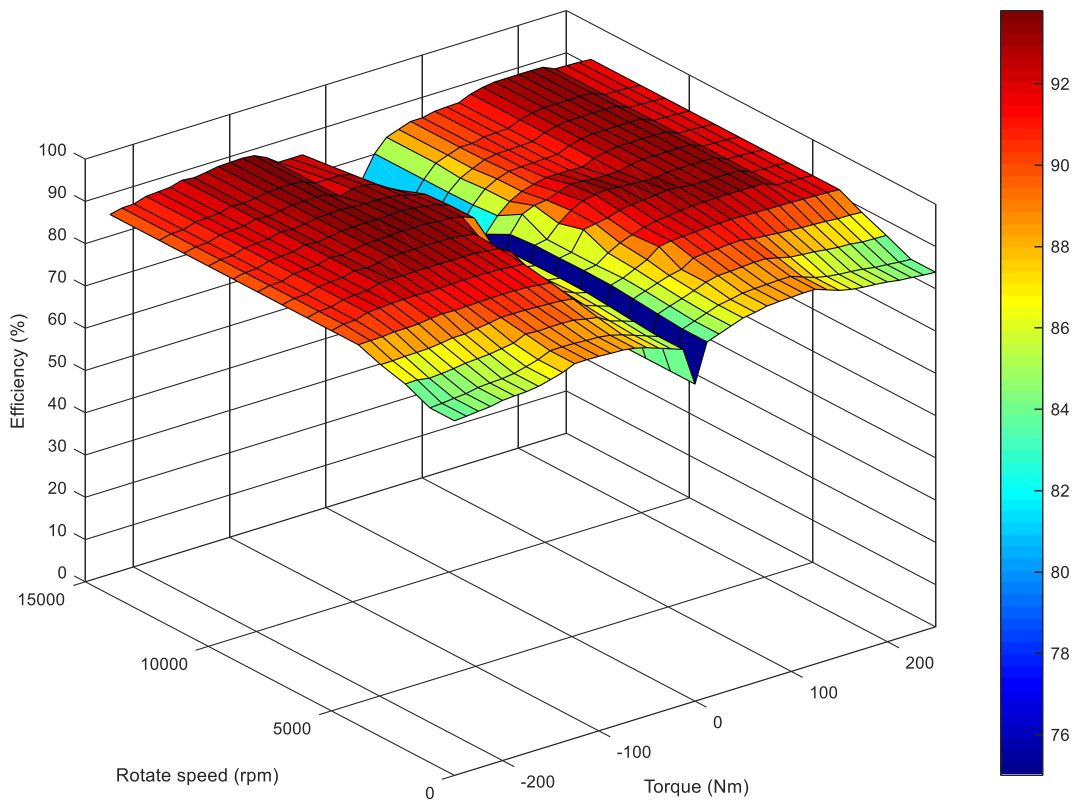

Fig. 2. Electric motor efficiency map.

$$T_m = \begin{cases} T_w \Big/ \left(\eta_{fd} \cdot R_{fd}\right), T_w \geq 0 \\ \eta_{fd} \cdot T_w \Big/ R_{fd}, T_w < 0 \end{cases} \tag{3}$$

$$w_m = w_w \cdot R_{fd} \tag{4}$$

The power demand is calculated as follows:

$$P_t = \begin{cases} T_m \cdot w_w / \eta_m, T_m \geq 0 \\ \eta_m \cdot T_m \cdot w_w, T_m < 0 \end{cases} \tag{5}$$

In the above, η_m is the motor efficiency, and can be found using an efficiency map of the motor, as shown in Fig. 2. R_{fd} is the final drive gear ratio.

The total power demand includes the power consumed by the auxiliary equipment, and is calculated as follows:

$$P_{req} = P_t + P_{aux} \tag{6}$$

The relationship correlating the fuel cell net power P_{fc}, battery power P_{bat}, and the total demanded power P_{req} is expressed by Eq. (7).

$$P_{fc} \cdot \eta_{DC/DC} + P_{bat} = P_{req} \tag{7}$$

2.2. Fuel cell modelling

The FCS is composed of two single stacks in parallel and has a maximum gross output power of 82 kW. Fig. 3(a) illustrates the relationship between the FCS net power and hydrogen consumption; the data was obtained through experiments at Tongji University. In the calculation, the hydrogen consumption is obtained by interpolation, in units of g s^{-1}.

The efficiency of the FCS is calculated based on the FCS net power and corresponding hydrogen consumption, as shown below.

$$M_{H_2} = \frac{1}{E_{low,H_2}} \int \frac{P_{fc}}{\eta_{fc}\left(P_{fc}\right)} dt \tag{8}$$

Here, M_{H_2} represents the hydrogen consumption (unit: g); E_{low,H_2} is the hydrogen lower heating value (120 kJ g^{-1}); and $\eta_{fc}(P_{fc})$ represents the FCS efficiency. The FCS net power and FCS efficiency curves are shown in Fig. 3(b).

Fuel cell degradation is mainly caused by unfavourable conditions, e.g. load changing cycles, start-stop cycles, idling, and high-power load conditions. Based on previous work [1], discrete expressions for the fuel cell degradation model are shown in Eqs. (9) and (10).

$$D_{fc} = \sum_{i=1}^{n}\left(d_{start-stop}(i) + d_{low}(i) + d_{load\_change}(i) + d_{high}(i)\right) \quad (9)$$

$$\begin{cases} Signal_{engine\_on}(i-1) = 0, Signal_{engine\_on}(i) = 1, d_{start-stop}(i) = 1.96 \times 10^{-3}, else\ d_{start-stop}(i) = 0 \\ 0 \le P_{fc}(i) < P_{low}\ and\ Signal_{engine\_on}(i) = 1,\ d_{low}(i) = 1.26 \times 10^{-3} \times \frac{\Delta t}{3600}, else\ d_{low}(i) = 0 \\ d_{load\_change}(i) = 5.93 \times 10^{-5} \times \frac{\left|P_{fc}(i) - P_{fc}(i-1)\right|}{\left(P_{high} - P_{low}\right)} \\ P_{fc}(i) > P_{high}, d_{high}(i) = 1.47 \times 10^{-3} \times \frac{\Delta t}{3600}, else\ d_{high}(i) = 0 \end{cases} \quad (10)$$

In the above, D_{fc} (%) represents the total performance degradation of the PEMFC; $d_{start-stop}, d_{low}, d_{load\_change}, d_{high}$ are the performance degradations caused by start-stop cycles, low-power load conditions, load changing cycles, and high-power load conditions at the ith time step, respectively; n is the number of time steps, and Δt is the length of a time step. $Signal_{engine\_on}$ represents the 'power on' signal of the FCS at the ith time step ('0' indicates shutdown, '1' indicates normal operation). P_{low} is the low-power threshold of the fuel cell under idling conditions, and P_{high} is the high-power threshold. Both P_{low} and P_{high} are determined based on the recommended operating voltage range of the PEMFC.

2.3. Battery modelling

The vehicle uses a lithium ion battery as the auxiliary power source. The relationship curves between the open-circuit voltage, charging and discharging resistance, and SOC of the battery at 25 °C are provided by the supplier, and are shown in Fig. 3(c), (d), and (e), respectively.

Based on the 'RINT' model [24], the relationships between the battery power, SOC, and current are shown in Eqs. (11) and (12).

$$P_{bat} = U_{oc}I_{bat} - I_{bat}^2 R_{bat} \quad (11)$$

$$SOC\left(t_f\right) = SOC(t_0) - \eta_c \frac{\int_{t_0}^{t_f} I_{bat}dt}{C_{bat}} \quad (12)$$

Here, C_{bat} is the capacity of the battery (unit: Ah), and η_c represents the coulomb efficiency of the battery, which is assumed to be 1 for a Li-ion battery [25].

The degradation of a battery is affected by various parameters, such as the charge and discharge cut-off voltages, SOC, temperature, charge and discharge rates, and current fluctuation frequency [26]. Based on Dakin's degradation approach for a battery [27], the aging rate increases exponentially with increasing values of the temperature, state of charge, and/or current magnitude. The discrete degradation model for the battery is expressed as shown in Eq. (13).

$$D_{bat} = \frac{\Delta t}{864}\sum_{i=1}^{n}\left[\exp\left(-\frac{b}{aT} + \frac{c \cdot SOC(i)}{a} + \frac{d}{a}\right) \times \exp\left(I_{bat}(i) \cdot \exp\left(\frac{e}{R_{bat}(i)T} + f\right)\right)\right] \quad (13)$$

In the above, D_{bat}(%) represents the total performance degradation of the battery; a, b, c, d, e, f are fitting coefficients; and T is the temperature (unit: K).

2.4. Daily driving cycle design

For the subsequent simulation, two 'three-day driving cycles' suitable for daily driving are designed. They are the 'three days driving cycle based on China_city' (hereinafter referred to as 'China_city') and 'three days driving cycle based on highway fuel economy test' (hereinafter referred to as HWFET), as shown in Fig. 4. The China_city cycle has a lower average speed, and its daily driving schedule is based on China_city for 0.662 h (two China-city driving cycles, total of 2386 s, 15.44 km), as follows: to work → parking for 8 h → driving 0.662 h back home → parking for 14.676 h. Likewise, the daily driving schedule for the HWFET comprises: driving based on HWFET for 0.425 h (two HWFET driving cycles, total of 1532 s, 33.02 km) to work → parking for 8 h → driving 0.425 h back home → parking 15.15 h.

3. Suboptimal real-time energy management strategy (EMS) formulation

3.1. Pontryagin's minimum principle (PMP) analysis and online implementation

PMP was derived by former Soviet Union researcher Lev Pontryagin. It is an optimal control algorithm. Based on PMP, the formulation of a Lagrange-type optimal control problem under the conditions of fixed time and a fixed end can be described as follows. P_{fc} is defined as the control variable of the system; SOC is the state variable; λ is the co-state; t_0 is the start time; and t_f is the end time. The control problem is to find the optimal control variable $P_{fc}^*(t) \in P_{fc}, t \in [t_0, t_f]$ enabling the controlled system shown in Eq. (14) to transfer from the given initial state to the terminal state, and to minimise the performance index function, as defined in Eq. (15).

$$\dot{SOC} = -\eta_c \frac{I_{bat}}{C_{bat}} \quad (14)$$

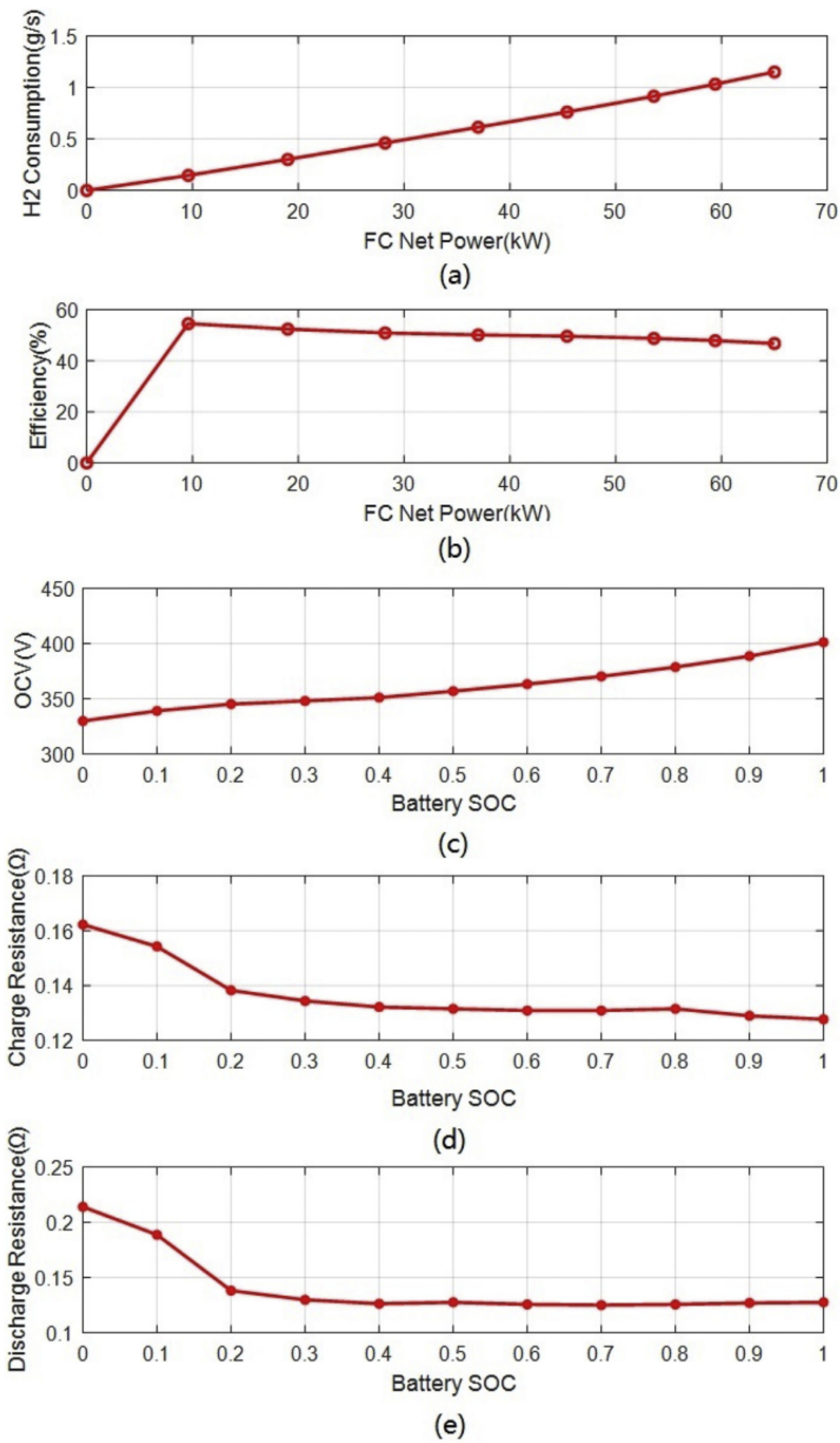

Fig. 3. Characteristics of the power sources used in this research: (a) relationship between net power of the fuel cell (FC) and hydrogen consumption, (b) efficiency curve of the FC system (FCS), (c) open circuit voltage of the battery at 25 °C, (d) charge resistance of the battery at 25 °C and (e) discharge resistance of the battery at 25 °C.

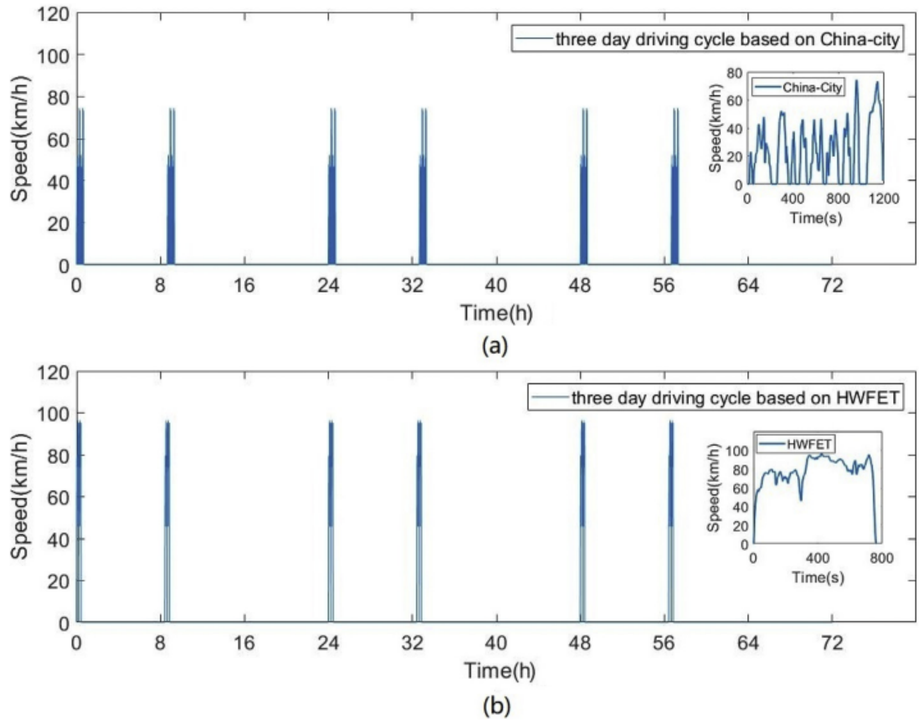

Fig. 4. (a) China_city driving cycle, (b) highway fuel economy test (HWFET) driving cycle.

$$J = \int_{t_0}^{t_f} \dot{m}\left(P_{fc}(t), t\right) dt \tag{15}$$

Here, $\dot{m}$ represents the hydrogen consumption rate (unit: g s^{-1}).

Based on PMP, the optimal FCS output power P_{fc}^* at each time step satisfies the following formula.

$$P_{fc}^* = \text{argmin } H \tag{16}$$

H is the Hamiltonian function, and is defined in Eq. (17), as follows:

$$H = \dot{m} - \lambda \cdot \dot{SOC} = \dot{m} - \lambda \cdot \eta_c \frac{U_{oc} - \sqrt{U_{oc}^2 - 4R_{bat}\left(P_{req} - P_{fc} \cdot \eta_{DC/DC}\right)}}{2R_{bat} \cdot C_{bat}} \tag{17}$$

In the above, λ is the co-state, and its state function is expressed as in Eq. (18) [28,29].

$$\dot{\lambda} = -\frac{\partial H}{\partial SOC} = \frac{\lambda \cdot \eta_c}{C_{bat}} \left(\frac{\partial I_{bat}}{\partial U_{oc}} \frac{\partial U_{oc}}{\partial SOC} + \frac{\partial I_{bat}}{\partial R_{bat}} \frac{\partial R_{bat}}{\partial SOC} \right) \tag{18}$$

Based on the above analysis, once the initial value of λ is determined, Eq. (19) can be solved, and the optimal P_{fc}^* can be obtained.

In the FCHEV problem, the co-state can be interpreted as the 'weight' coefficient of the $\dot{SOC}$, and the second term in Eq. (17) can be considered as the equivalent fuel consumption [30]. When the battery SOC is in a certain range (e.g. between 0.5 and 0.7), the voltage and resistance hardly change with the SOC, as shown in Fig. 3. Thus, according to Eq. (18), $\dot{\lambda} \approx 0$, and λ can be considered as a constant [31–33].

In PMP-based discrete optimal control, the optimal P_{fc}^* is obtained by solving Eq. (16) at every time step, under the system constraints shown in Eq. (19).

$$S \cdot t \begin{cases} P_{fc,\min} \le P_{fc} \le P_{fc,\max} \\ P_{bat,\min} \le P_{bat} \le P_{bat,\max} \\ SOC_{\min} \le SOC \le SOC_{\max} \end{cases} \tag{19}$$

According to the above analysis, a series of optimal control simulations based on PMP are conducted with a fixed initial SOC ($SOC(t_0)$) and different constant values of co-states under different driving cycles, and the final SOC ($SOC(t_f)$) values are recorded separately. The relationships between the curves of the battery ΔSOC and co-states in different driving cycles are shown in Fig. 5(a), where $\Delta SOC = SOC(t_f) - SOC(t_0), SOC(t_0) = 0.6$. The driving cycles in these simulations include low-average-speed driving cycles, such as 'idling 0 km/h', 'New European Driving Cycle', and 'Urban Dynamometer Driving Schedule' (UDDS), as well as high-average-speed driving cycles, such as the HWFET, 'US06 Supplemental Federal Test Procedure' (US06), 'US06 Highway sections', and '120 km/h cruising'. These driving cycles can represent almost all of the daily driving scenarios for passenger vehicles in China.

Fig. 5 illustrates that the relationship between the ΔSOC and co-state is monotonic, indicating that the battery SOC can be regulated through regulation of the co-state. When the battery SOC is low, a smaller co-state can be used, so that the battery SOC is continuously rising (even at high-speed driving cycles). When the battery SOC is high, a larger value of the co-state can be used, so that the battery SOC continuously drops (even at a low-speed driving cycle). Thus, according to the battery SOC, the value of the co-state can be updated in real time, thereby realising online application of PMP.

To make this strategy applicable to any driving cycle, it is necessary to obtain a general curve, by synthesising the curves under each typical driving cycle. The typical conditions are divided into two groups, i.e. low-speed cycles and high-speed cycles, as shown in Fig. 5(a). The curves are weighted after averaging within the group. As the vehicles investigated in this study are mainly used for urban commuting, low-speed conditions are more common.

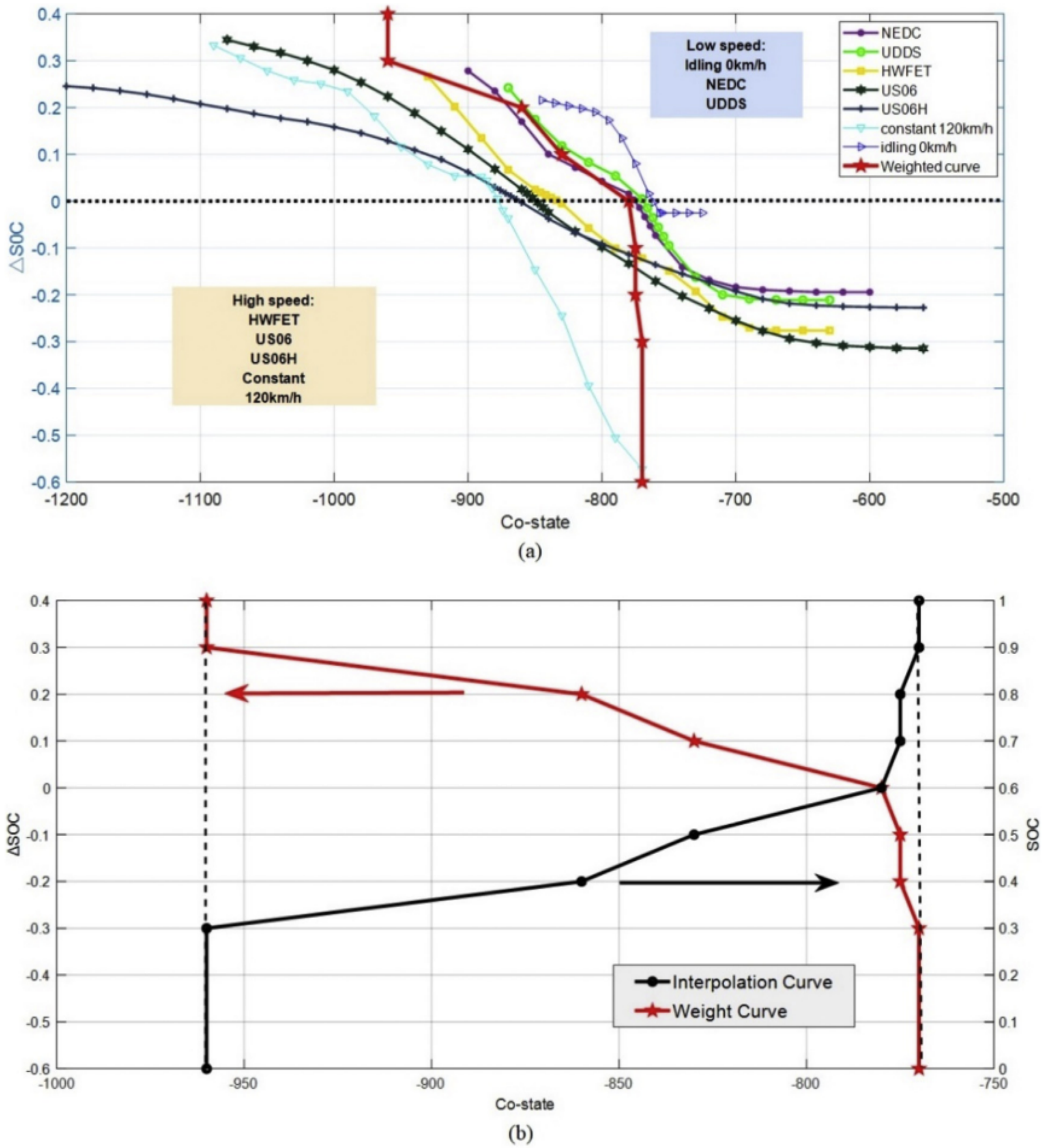

Fig. 5. (a)The relationship between battery state of charge (ΔSOC) and constant co-state in different driving cycles and(b) interpolation curve of the SOC and co-state.

Therefore, the weight factor for the low-speed group was set as 0.8. The red line in Fig. 5(a) is the weighted curve. The SOC of the battery should be maintained within the range of 0.3–0.9 to ensure normal driving and braking energy recovery. According to the upper and lower bounds of the SOC, the boundary of the co-state variable value can be determined.

By ignoring the influence of the initial SOC on the strategy, it is assumed that the same co-state will lead to the same ΔSOC under different initial SOCs. Based on the above assumptions, a coordinate transformation is conducted for the weighted curve. In general, the SOC of the battery is expected to be maintained at approximately 0.6, to minimise driver anxiety owing to the short driving range of pure electricity, and to slow the battery degradation. The battery SOC can be obtained in real time at each time step, and is regarded as the initial SOC. The final SOC is set as 0.6, so that the mapping between the current SOC and ΔSOC can be established. Then, the interpolation curve between the SOC and co-state can be obtained, as shown in Fig. 5(b).

By applying the interpolation curve to a control process, a PMP-based EMS is developed. In each time step, the controller obtains the current SOC of the battery, and the value of the co-state is determined based on the SOC. Then, the optimal P_{fc}^{*} is solved for, and minimises the Hamiltonian function. The strategy does not depend on prior knowledge of the driving cycles. In addition, the solution for the minimum value of the Hamiltonian function does not require significant computational resources, so the proposed EMS can be easily applied in real vehicles.

The EMS is verified under the UDDS and US06 driving cycles. The simulation results in Fig. 6 show that the final value of the SOC is controlled within a certain range (based on the premise that the

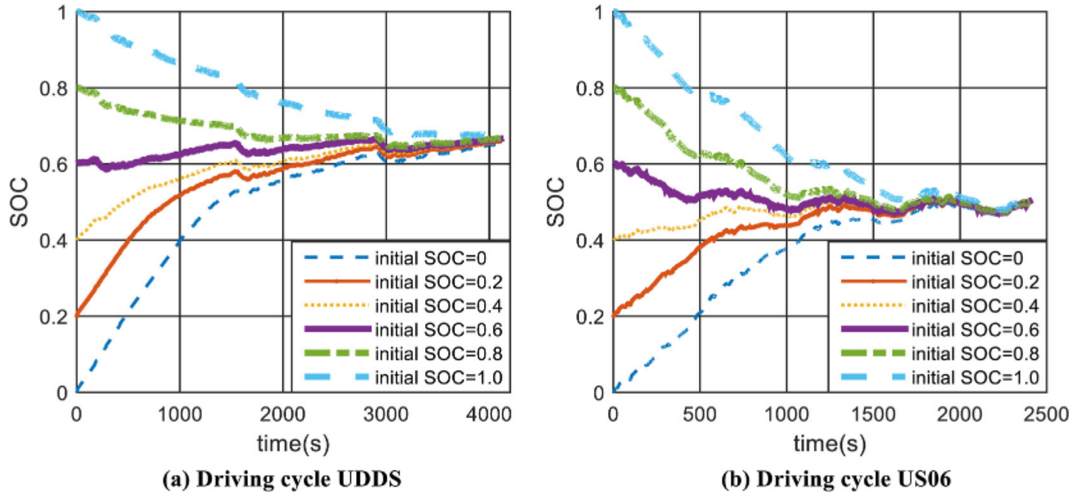

Fig. 6. Change of SOC under control of the proposed method in different driving cycles.

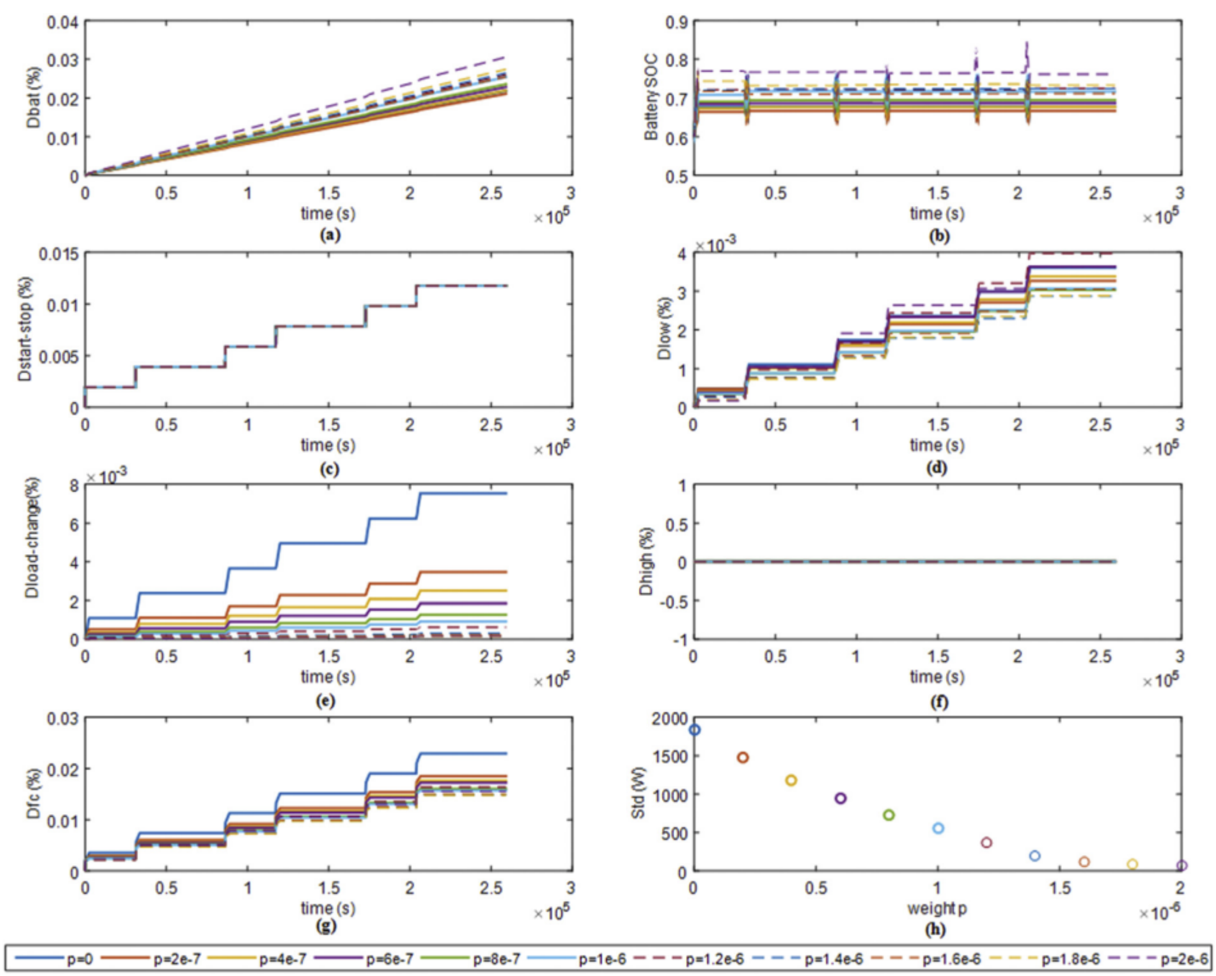

Fig. 7. Degradation of power sources under driving cycle China_city.

initial SOC is any value between 0 and 1), proving that the online updating method for the co-state is effective.

3.2. EMS considering both fuel economy and power source durability

In Subsection 3.1, a real-time EMS was developed. However, this EMS only realises an approximate optimal fuel economy, and does

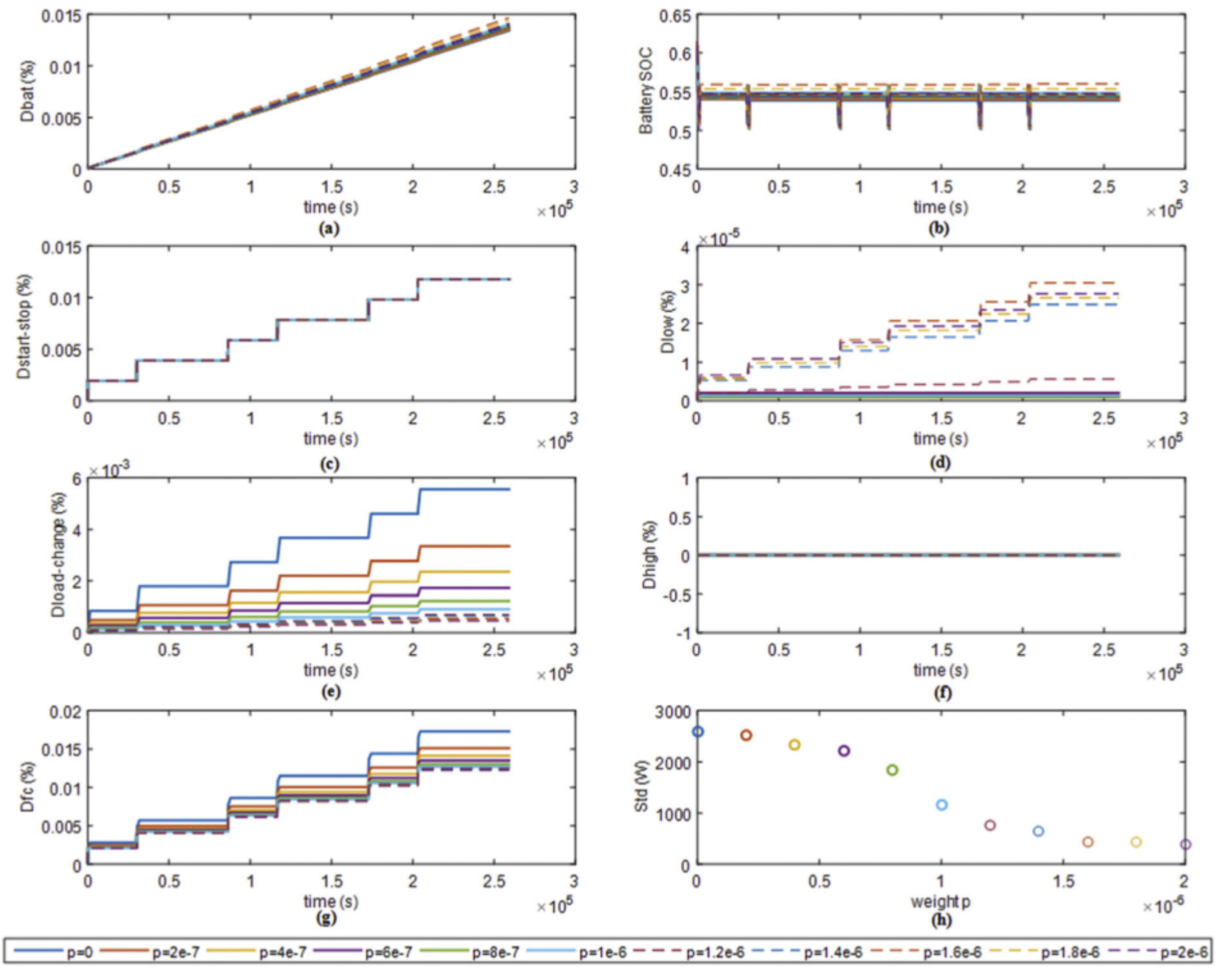

Fig. 8. Degradation of power sources under driving cycle HWFET.

not consider the durability of power sources. As mentioned above, the main cause of degradation in fuel cells is excessive changes in power. To suppress the power changes of the fuel cell, a power variation limiting factor $\left|P_{fc}(t) - P_{fc}(t - \Delta t)\right|$, serving as a penalty item, is added to the performance index function with a weight coefficient p, as shown in Eq. (20). Then, the new Hamiltonian function is defined as shown in Eq. (21).

$$J = \sum_{i=1}^{n}\left(\dot{m}\left(P_{fc}(i)\right) \cdot \Delta t + p \cdot \left|P_{fc}(i) - P_{fc}(i-1)\right|\right) \tag{20}$$

$$H = \dot{m} - \lambda \cdot \dot{SOC} + p \cdot \left|P_{fc}(i) - P_{fc}(i-1)\right| \tag{21}$$

In Subsection 3.1, the online co-state updating method was provided; thus, the weight p is the last parameter to be determined. Under the two types of daily driving cycles introduced in Subsection 2.4, the suppression of the fuel cell output power variation and deterioration of battery operating status under different weights p are analysed through simulations. The goal is to determine a weight p that balances the fuel economy and power source durability.

The battery and fuel cell degradation with different p values under the two driving cycles are shown in Figs. 7 and 8. In general, the degradation of the fuel cell tends to decrease with increases of p, mainly because the power variation of the fuel cell is suppressed. However, there are some differences between the simulation results, owing to the characteristics of the driving cycles. The China_city driving cycle has a relatively low average bus-demanded power (for the vehicle studied in this work, approximately 4.6 kW), and the output power of the fuel cell more frequently falls under the low power threshold. Therefore, the total degradation of the fuel cell mainly depends on the degradation caused by power changes and low power (there is no difference in the degradation caused by start/stop and high power with different p). As there is no special control for the low power output of the fuel cell, the degradation of fuel cell shows a certain degree of irregularity. In contrast, the HWFET cycle has a relatively high average bus-demanded power (approximately 16.6 kW), and the fuel cell degradation is mainly influenced by power changes (the degradation caused lower power is two orders of magnitude lower than the degradation caused by power changes). Although the degradation caused by the low power varies irregularly with the increase of p, the total degradation decreases.

Battery degradation is mainly influenced by the battery SOC. Owing to the low power demand in the China_city cycle, the battery SOC is maintained at a high level; thus, the battery degradation is higher. In addition, the decrease of fuel cell output power change leads to the increase of battery output power change, and the battery degradation is correspondingly increased.

Finally, the standard deviation of the fuel cell output power is calculated as shown in Eq. (22), where Std is the average of the standard deviation of P_{fc} every 50 time steps. The value of Std reflects the dispersion of the 50 power output points relative to their

Table 2
Fuel consumption for different weights p.

Driving cycle	p	m(g)	Initial SOC	End SOC	m_e(g)	M(g)
China_city (Distance: 92.7 km; time: 259200 s)	0	1043.33	0.6	0.6777	60.25	**983.08**
	2e-7	1036.29	0.6	0.6668	51.75	984.54
	4e-7	1048.50	0.6	0.6770	59.67	988.84
	6e-7	1059.54	0.6	0.6866	67.14	992.40
	8e-7	1070.51	0.6	0.6951	73.89	996.62
	1e-6	1090.10	0.6	0.7171	91.19	998.92
	1.2e-6	1099.17	0.6	0.7248	97.20	1001.98
	1.4e-6	1106.44	0.6	0.7242	96.76	1009.68
	1.6e-6	1097.17	0.6	0.7114	86.64	1010.53
	1.8e-6	1112.04	0.6	0.7329	103.65	1008.40
	2e-6	1136.27	0.6	0.7614	126.08	1010.19
Highway fuel economy test (HWFET) (Distance: 198.1 km; time: 259200 s)	0	2320.89	0.6	0.5386	47.06	**2367.95**
	2e-7	2322.93	0.6	0.5398	46.03	2368.96
	4e-7	2327.10	0.6	0.5416	44.71	2371.82
	6e-7	2331.04	0.6	0.5433	43.39	2374.43
	8e-7	2333.83	0.6	0.5451	42.07	2375.90
	1e-6	2339.06	0.6	0.5485	39.44	2378.49
	1.2e-6	2340.60	0.6	0.5470	40.61	2381.21
	1.4e-6	2340.05	0.6	0.5430	43.69	2383.73
	1.6e-6	2352.17	0.6	0.5601	30.64	2382.81
	1.8e-6	2347.82	0.6	0.5537	35.48	2383.29
	2e-6	2343.09	0.6	0.5477	40.17	2383.26

average value, i.e. the fuel cell output power fluctuation level. As shown in Figs. 7(h) and b8(h), as the weight increases, Std decreases, and the fluctuation of P_{fc} becomes smaller.

$$Std = \left(\sum_{i=1}^{N} \sqrt{\frac{\sum_{50(i-1)+1}^{50i} \left(P_{fc} - \overline{P}_{fc}(i) \right)^2}{50}} \right) \Big/ N \tag{22}$$

The fuel consumptions for different weights p under the two driving cycles (China_city and HWFET) are shown in Table 2, where m represents the fuel consumption during the simulation; and m_e represents the equivalent hydrogen consumption, i.e. the amount of hydrogen consumed (or saved) when the battery is charged (or discharged) from the end SOC to the initial SOC. m_e is calculated as shown in Eq. (23). M represents the real hydrogen consumption, and is equal to m plus m_e (or m minus m_e).

$$m_e = \frac{E \times 3.6 \times 10^6}{E_{low,H_2} \times \eta_{fc}} \tag{23}$$

E represents the energy capacity of the battery (unit: kWh).

The hydrogen consumption data in Table 2 indicate that under the two daily driving cycles, the real hydrogen consumption M increases with increasing p. This is because the larger the weight, the farther the calculation result deviates from the optimal control when only considering the fuel economy.

The above simulation results contain information such as the fuel cell degradation, battery degradation, and hydrogen consumption. To determine p, it is necessary to design an evaluation function capable of comprehensively reflecting and evaluating the fuel economy and power source durability of the FCHEV, as shown in Eq. (24).

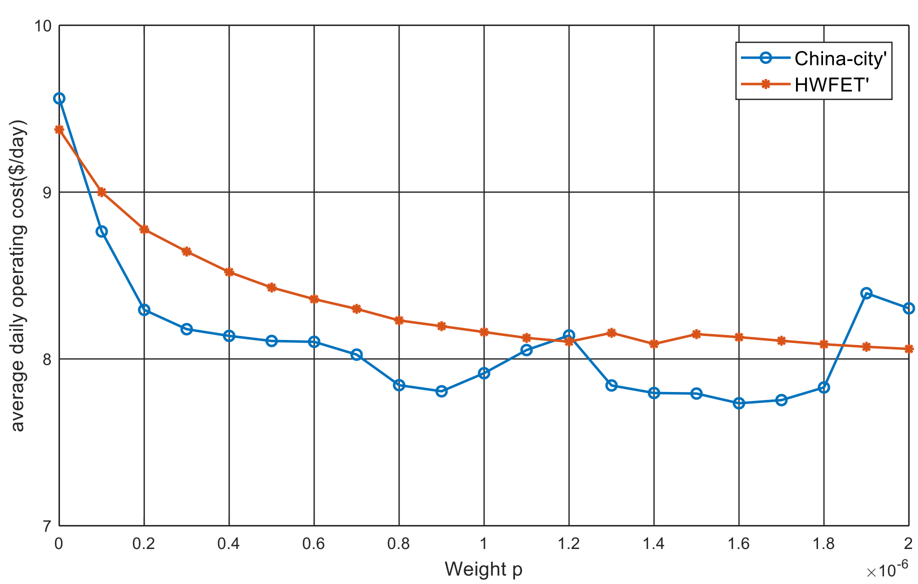

Fig. 9. The average daily operating cost with different p.

$$Cost = \left(\frac{D_{fc}}{10} \times c_{fc} \times P_{fcmax} + \frac{D_{bat}}{20} \times c_{bat} \times C_{batmax} + \frac{M}{1000} \times c_{H_2} \right) \tag{24}$$

In the above, *Cost* represents the average daily operating cost of the FCHEV ($), and does not include consumption from the accessories system and other components, such as the consumption of humidified water. c_{fc} (\$ kW^{-1}), c_{bat} (\$/kWh), and c_{H_2}(\$ kg^{-1}) are the prices of the fuel cell, battery, and hydrogen, respectively; $P_{fc\max}$ is the maximum output power of the fuel cell, and in our work, $P_{fc\max} = 82$ kW; and $C_{bat\max}$ is the maximum capacity of the battery. In our work, $C_{bat\max} = 13$ kWh; '10' (%) and '20' (%) are the allowed performance degradation percentages for the FC and battery, respectively.

The evaluation function consists of three parts. The first part comprises evaluating the equivalent daily average cost of the fuel cell degradation. According to the U.S. DOE's investigation and forecasts of fuel cell costs [34], the cost of sample fuel cells is approximately 1826 \$ kW^{-1}; the cost for mass production is approximately 100 \$ kW^{-1}, and the target cost is 50 \$ kW^{-1}, indicating that a possible c_{fc} is among them. The second part comprises evaluating the equivalent daily average cost of battery degradation. The current cost of EV batteries is approximately 250–500 \$ kWh^{-1}, and the target cost in a few years is approximately 175–350 \$ kWh^{-1} [35]. The third part of the evaluation concerns the daily cost of hydrogen consumption. According to data from the U.S. DOE [36], the current cost of hydrogen is approximately 4.4 \$ kg^{-1}, and its target is 2 \$ kg^{-1}. According to Eq. (24), the evaluation results for different p values are illustrated in Fig. 8, where $c_{fc} = 100$ \$ kWh^{-1}, $c_{bat} = 400$ \$ kWh^{-1}, and $c_{H_2} = 4.4$\$ kg^{-1}.

As shown in Fig. 9, the *Cost* under the HWFET driving cycle decreases with increasing p values. In contrast, under the China_city driving cycle, and owing to irregular idling and low load control of the fuel cell, when p increases, the average daily operating cost becomes subject to sudden increases. The vehicle investigated in this study is a passenger car mainly used in cities and driving at medium-slow speeds is more common than at high speeds. Thus, $p = 1.6\text{e-}6$ is ultimately selected, as it is the minimum point of the China_city cost curve.

4. Contrast with other EMSs

The EMS based on the approximate PMP is formulated after determining the co-state λ and weight p. The simulation results for power source degradation and hydrogen consumption under control of the proposed EMS are provided and analysed. Although the rule-based EMS is not ideal in terms of fuel economy, it is relatively simple and requires less computation, so it can be well-applied to real vehicles [37]. DP is a global optimisation algorithm that can obtain global optimal results, and is often used as a benchmark for other EMSs [37]. For further confirmation of the superiority of the proposed EMS, a comparison between the proposed EMS and two other EMSs is provided. The first EMS is a rule-based power-following EMS (RB); its basic control concept is that the demanded power is mainly provided by the FCS, i.e.$P_{fc} = P_{req}$, with any insufficiency being supplemented by the auxiliary power source. The second EMS is based on the DP, with the objective of minimising the average daily operating cost.

The results for the different EMSs, including the fuel cell degradation, battery degradation, hydrogen consumption, and average daily operating cost, are shown in Table 3. The proposed strategy reduces fuel cell degradation, including the degradation caused by dynamic loading, as well as that caused by idling and low load conditions. Compared with the RB under the China_city cycle, the PMP-based strategies sacrificed battery durability and hydrogen consumption, but the reduction of the fuel cell degradation was significant. Compared with DP-based EMS, the proposed PMP-based EMS has nearly the same average daily operating cost, indicating that the proposed EMS is approximately optimal. It should be emphasised that the proposed EMS realises an online implementation of PMP, which is of great significance to engineering applications. In general, the PMP-based EMS had a sub-optimal average daily operating cost and offered comprehensively superior fuel economy and power source durability, under both the low-speed urban driving cycle and high-speed suburban driving cycle.

5. Conclusions

The paper has presented an approximately optimal PMP-based EMS for FCHEVs, considering both fuel cell economy and power source durability. First, degradation models are introduced for the power sources. Second, a co-state updating method is provided according to the battery SOC, and a fuel cell power variation limiting factor is added into the PMP with a weight coefficient, in consideration of power source durability. Finally, two three-day driving cycles named China_city (three-day driving cycle based on China_city) and HWFET (three-day driving cycle based on HWFET) suitable for the daily driving of vehicles are designed for evaluating the proposed EMS. The simulation results show that the proposed EMS performs well in various daily driving conditions, e.g. in regard to the fuel cell degradation caused by starting and stopping, idle/low-power, dynamic loading and high-power output, battery degradation, and average daily operating cost. An evaluation function considering both hydrogen consumption and power source degradation is used to comprehensively assess the EMS. A comparison with a DP-based strategy shows that the PMP-based EMS is approximately optimal. Above all, by establishing the

Table 3
Comparison between the proposed energy management strategy (EMS) and two other EMSs.

Driving cycles		China_city					HWFET				
EMSs		Rule-based approach (RB)	Dynamic programming (DP)	Pontryagin's minimum principle (PMP)	PMP cp. with RB	PMP cp. with DP	RB	DP	PMP	PMP cp. with RB	PMP cp. with DP
$D_{fc} \times 10^{-3}$ (%)	$d_{start\text{-}stop}$	11.7600	11.7600	11.7600	→	→	11.7600	11.7600	11.7600	→	→
	d_{low}	3.9543	1.9246	3.0506	↓22.9%	↑58.5%	0.8747	0	0.0305	↓96.5%	/
	d_{load_change}	18.7230	0.1779	0.2127	↓98.9%	↑19.6%	16.5793	0.5414	0.5517	↓96.7%	↑1.90%
	d_{high}	0	0	0	→	→	0	0	0	→	→
	D_{fc}	34.4373	13.8625	15.0233	↓56.4%	↑8.37%	29.2140	12.3014	12.3422	↓57.8%	↑3.32%
$D_{bat} \times 10^{-3}$ (%)		23.7269	28.3119	25.4125	↓7.1%	↓10.2%	20.5668	13.4372	14.6481	↓28.8%	↑9.01%
H_2 consumption(g)		995.78	982.55	1010.53	↑1.5%	↑2.85%	2388.99	2299.37	2382.81	↓0.3%	↑3.6%
*Cost** (\$ day^{-1})		12.9297	7.6839	7.7909	↓39.7%	↑1.39%	13.2715	8.0043	8.1378	↓38.7%	↑1.67%

Note: * where $c_{fc} = 100\$ \text{ kW}^{-1}$; $c_{bat} = 400\$ \text{ kWh}^{-1}$; $c_{H_2} = 4.4\$ \text{ kg}^{-1}$.

interpolation curve, the co-state can be updated online according to the battery SOC, ensuring that the PMP-based EMS is not dependent on the prior knowledge of driving cycles. As such, it can be applied in real-time control. In addition, the results of the evaluation function provide directions for further optimisation of the EMS.

Although a lot of work has been done, there are still some limitations and challenges: 1) In this paper, the economy of FCHEV is first optimized, and then the durability of power source is considered. Considering both fuel economy and power source durability at the same time can get better results, but also more challenging; 2) The strategy proposed in this paper can be verified by experiments to enhance the reliability of the results. In the future, we will conduct further research on these limitations and challenges.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Ke Song: Conceptualization, Validation, Writing - review & editing, Project administration. **Xiaodi Wang:** Investigation, Methodology, Software, Writing - original draft. **Feiqiang Li:** Funding acquisition, Resources. **Marco Sorrentino:** Writing - review & editing. **Bailin Zheng:** Supervision.

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References

[1] Song K, Chen H, Wen P, Zhang T, Zhang B, Zhang T. A comprehensive evaluation framework to evaluate energy management strategies of fuel cell electric vehicles. Electrochim Acta 2018;292:960–73.
[2] Zhang T, Pota H, Chu C, Gadh R. Real-time renewable energy incentive system for electric vehicles using prioritization and cryptocurrency. Appl. Energy 2018;226:582–94.
[3] Sulaiman N, Hannan MA, Mohamed A, Ker PJ, Majlan EH, Wan Daud WR. Optimization of energy management system for fuel-cell hybrid electric vehicles: issues and recommendations. Appl. Energy 2018;228:2061–79.
[4] Tang Y, Yuan W, Pan M, Wan Z. Experimental investigation on the dynamic performance of a hybrid PEM fuel cell/battery system for lightweight electric vehicle application. Appl. Energy 2011;88:68–76.
[5] Feroldi D, Carignano M. Sizing for fuel cell/supercapacitor hybrid vehicles based on stochastic driving cycles. Appl. Energy 2016;183:645–58.
[6] Zhou D, Ravey A, Al-Durra A, Gao F. A comparative study of extremum seeking methods applied to online energy management strategy of fuel cell hybrid electric vehicles. Energy Convers Manage 2017;151:778–90.
[7] Ettihir K, Higuita Cano M, Boulon L, Agbossou K. Design of an adaptive EMS for fuel cell vehicles. Int. J. Hydrogen. Energy 2017;42:1481–9.
[8] Simmons K, Guezennec Y, Onori S. Modeling and energy management control design for a fuel cell hybrid passenger bus. J Power Sources 2014;246:736–46.
[9] Chen Z, Mi CC, Xia B, You C. Energy management of power-split plug-in hybrid electric vehicles based on simulated annealing and Pontryagin's minimum principle. J Power Sources 2014;272:160–8.
[10] Ou K, Yuan W, Choi M, Yang S, Jung S, Kim Y. Optimized power management based on adaptive-PMP algorithm for a stationary PEM fuel cell/battery hybrid system. Int. J. Hydrogen. Energy 2018;43:15433–44.
[11] Wang Y, Wu Z, Chen Y, Xia A, Guo C, Tang Z. Research on energy optimization control strategy of the hybrid electric vehicle based on Pontryagin's minimum principle. Comput Electr Eng 2018;72:203–13.
[12] Schmid R, Bürger J, Bajcinca N. A comparison of PMP-based energy management strategies for plug-in-hybrid electric vehicles. IFAC-PapersOnLine 2019;52:592–7.
[13] Xu L, Ouyang M, Li J, Yang F, Lu L, Hua J. Application of Pontryagin's Minimal Principle to the energy management strategy of plugin fuel cell electric vehicles. Int. J. Hydrogen. Energy 2013;38:10104–15.
[14] Li X, Wang Y, Yang D, Chen Z. Adaptive energy management strategy for fuel cell/battery hybrid vehicles using Pontryagin's Minimal Principle. J Power Sources 2019;440:227105.
[15] Xie S, Hu X, Xin Z, Brighton J. Pontryagin's Minimum Principle based model predictive control of energy management for a plug-in hybrid electric bus. Appl. Energy 2019;236:893–905.
[16] Onori S, Tribioli L. Adaptive Pontryagin's Minimum Principle supervisory controller design for the plug-in hybrid GM Chevrolet Volt. Appl. Energy 2015;147:224–34.
[17] Pei P, Chen H. Main factors affecting the lifetime of Proton Exchange Membrane fuel cells in vehicle applications: a review. Appl. Energy 2014;125: 60–75.
[18] Li Q, Chen W, Liu Z, Li M, Ma L. Development of energy management system based on a power sharing strategy for a fuel cell-battery-supercapacitor hybrid tramway. J Power Sources 2015;279:267–80.
[19] Florescu A, Bacha S, Munteanu I, Bratcu AI, Rumeau A. Adaptive frequency-separation-based energy management system for electric vehicles. J Power Sources 2015;280:410–21.
[20] Lachhab I, Krichen L. An improved energy management strategy for FC/UC hybrid electric vehicles propelled by motor-wheels. Int. J. Hydrogen. Energy 2014;39:571–81.
[21] Zheng CH, Xu GQ, Park YI, Lim WS, Cha SW. Prolonging fuel cell stack lifetime based on Pontryagin's Minimum Principle in fuel cell hybrid vehicles and its economic influence evaluation. J Power Sources 2014;248:533–44.
[22] Liu C, Liu L. Optimal power source sizing of fuel cell hybrid vehicles based on Pontryagin's minimum principle. Int. J. Hydrogen. Energy 2015;40:8454–64.
[23] Li T, Liu H, Zhao D, Wang L. Design and analysis of a fuel cell supercapacitor hybrid construction vehicle. Int. J. Hydrogen. Energy 2016;41:12307–19.
[24] Li J, Wang Y, Chen J, Zhang X. Study on energy management strategy and dynamic modeling for auxiliary power units in range-extended electric vehicles. Appl. Energy 2017;194:363–75.
[25] Zheng Y, Gao W, Ouyang M, Lu L, Zhou L, Han X. State-of-charge inconsistency estimation of lithium-ion battery pack using mean-difference model and extended Kalman filter. J Power Sources 2018;383:50–8.
[26] Ouyang M, Feng X, Han X, Lu L, Li Z, He X. A dynamic capacity degradation model and its applications considering varying load for a large format Li-ion battery. Appl. Energy 2016;165:48–59.
[27] Baghdadi I, Briat O, Delétage J, Gyan P, Vinassa J. Lithium battery aging model based on Dakin's degradation approach. J Power Sources 2016;325:273–85.
[28] Kim NW, Lee DH, Zheng C, Shin C, Seo H, Cha SW. Realization of PMP-based control for hybrid electric vehicles in a backward-looking simulation. Int. J Auto. Tech-KOR 2014;15:625–35.
[29] Zhang J, Zheng C, Cha SW, Duan S. Co-state variable determination in Pontryagin's Minimum Principle for energy management of hybrid vehicles. Int. J. Precis Eng. Man 2016;17:1215–22.
[30] Kim N, Rousseau A, Lee D. A jump condition of PMP-based control for PHEVs. J Power Sources 2011;196:10380–6.
[31] Kim N, Cha S, Peng H. Optimal control of hybrid electric vehicles based on pontryagin's minimum principle. IEEE T CONTR SYST T 2011;19:1279–87.
[32] Ettihir K, Boulon L, Agbossou K. Optimization-based energy management strategy for a fuel cell/battery hybrid power system. Appl. Energy 2016;163: 142–53.
[33] Zheng C, Xu G, Xu K, Pan Z, Liang Q. An energy management approach of hybrid vehicles using traffic preview information for energy saving. Energy Convers Manage 2015;105:462–70.
[34] Marcinkoski J, Spendelow J, Wilson A, Papageorgopoulos D. Fuel cell system cost-2015. DOE Hydrogen and Fuel Cells Program Record; 2015.
[35] Nykvist B, Nilsson M. Rapidly falling costs of battery packs for electric vehicles. Nat Clim Change 2015;5:329–32.
[36] Parks G. Hydrogen station compression, storage, and dispensing technical status and costs. National Renewable Energy Laboratory; 2014.
[37] Peng J, He H, Xiong R. Rule based energy management strategy for a series–parallel plug-in hybrid electric bus optimized by dynamic programming. Appl. Energy 2017;185:1633–43.